(14)

1924260352

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE:Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I R. Hashini (1924260352) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Hash

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1.40	1.40	0.80
Concept Mapping Structure	25%	1.75	1.75	1.00
Linking & Relationships	20%	1.40	1.40	0.80
Application & Analysis	20%	1.40	1.40	0.80
Explanation & Clarity	15%	1.05	1.05	0.60

Signature of the Faculty

Total Marks 18 /30

CONCEPT MAPPING QUESTIONS

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept		Constructs a clear, logical, and well-	Concept map is	Concept map	No clear

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

18/30

Reg No: 1924260352
Name : R. Harshini
Course Name : Computer Networks
Course Code : CSA0709

Assessment tool - 2

Concept map

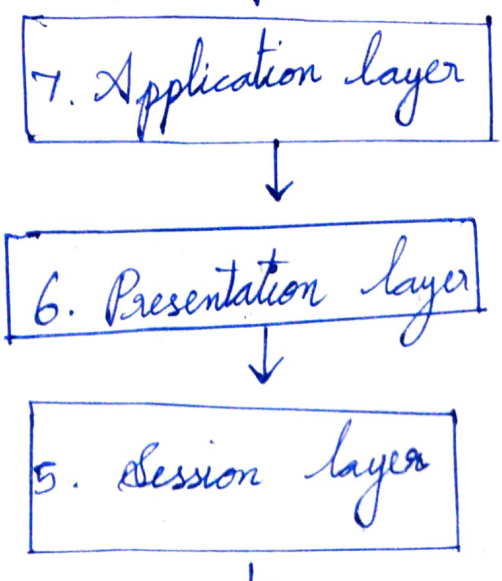
Questions :

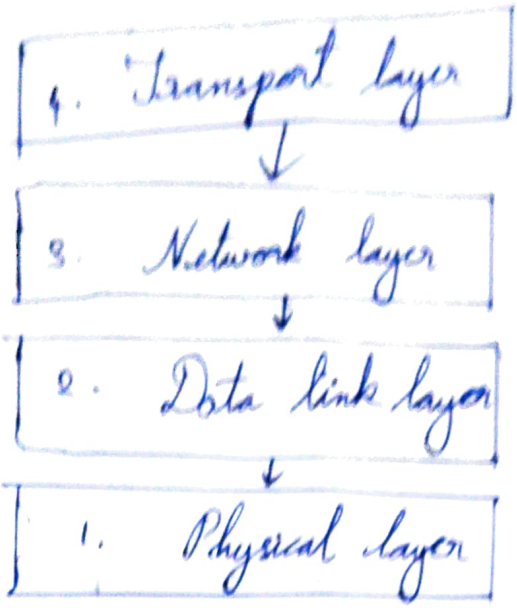
1. A partially completed concept map is given below :

Application → Presentation → _____ → Transport network → _____ → Physical. Analyze the missing layers and explain how their functions essential for reliable communication.

Answer :

1. OSI layers - Concept map





Missing answers:

1. Session layer
2. Data link layer

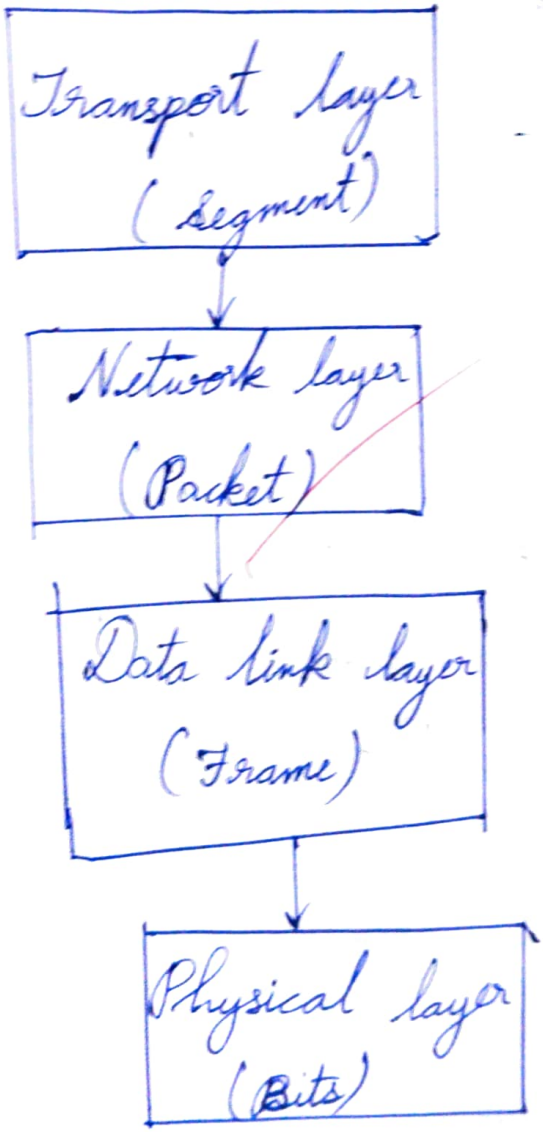
Functions and importance

- Session layer : Establishers, manages and terminates communication sessions between devices.
- Data link layer: Transfers frames between directly connected devices and detects transmission errors.
- Transport layer: Provides end to end delivery can support reliable data transfer.
- Network layer: Selects routes and forwards packets between networks.

clusion:

All seven OSI layers work together to help data travel from the sender to the receiver accurately and reliably.

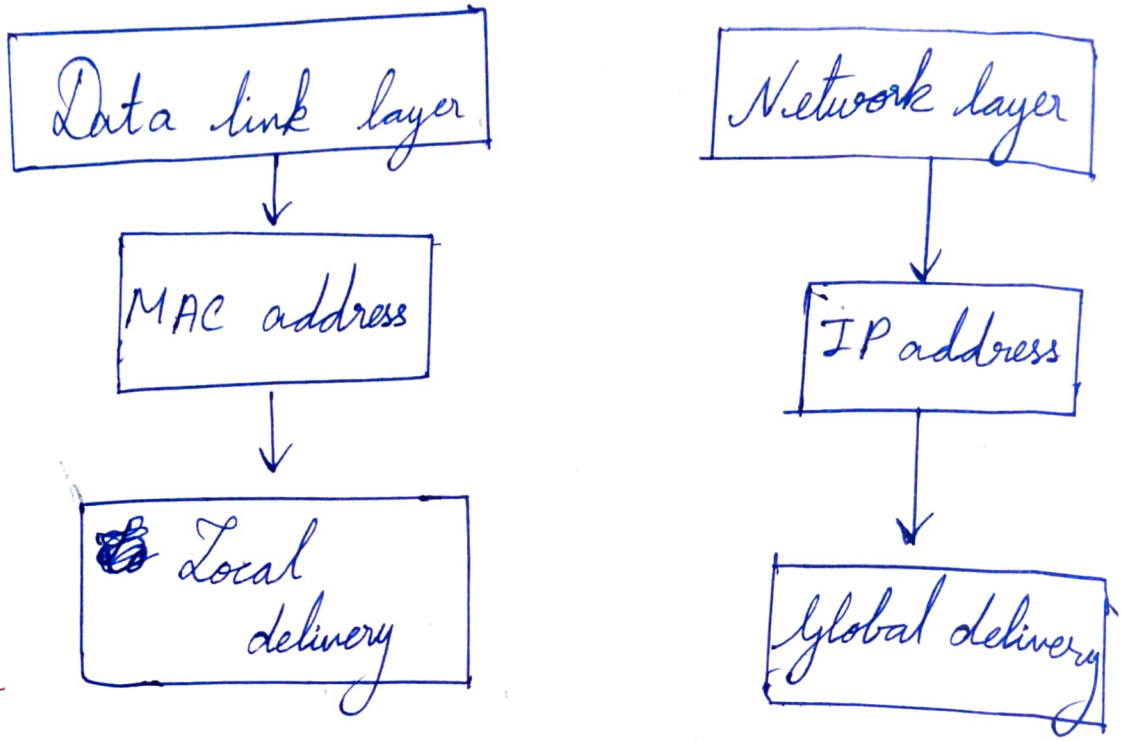
2. Data units and OSI layers - Concept map.



Conclusion:

Data transformation allows each OSI layer to

3 MAC address and IP address - Concept Map



Conclusion:

IP addresses support delivery across networks, while MAC addresses support delivery over individual local links. Together, they help data to reach the correct destination.